Report: Tooth Numbering Detection using YOLOv11

Methodology/Approach

- First the raw dataset containing 497 images along with corresponding labels which was in zip format was unzipped. Then it was converted into the train, test, validation format which was required to train the YOLO model using the reorganize-script.py file with 70% train, 20% val and 10% test.

- Upon analysis, I discovered the dataset's ground-truth labels were grouped by tooth type rather than spatial location. To correct this without manually relabelling 500 images, I remade the data.yaml to act as a translation layer, mapping the tooth types directly to their correct FDI standard coordinates. With the following mapping:

```
# Canines
0: '13'
1: '23'
2: '33'
3: '43'
# Central Incisors
4: '21'
5: '41'
6: '31'
7: '11'
# First Molars
8: '16'
9: '26'
10: '36'
11: '46'
# First Premolars
12: '14'
13: '34'
14: '44'
15: '24'
# Lateral Incisors
16: '22'
17: '32'
18: '42'
19: '12'
# Second Molars
20: '17'
21: '27'
22: '37'
23: '47'
# Second Premolars
24: '15'
25: '25'
26: '35'
```

27: '45'
Third Molars (Wisdom Teeth)
28: '18'
29: '28'
30: '38'
31: '48'

- Then I uploaded the dataset along with data.yaml to Kaggle and started writing the python script to train the model. I chose to train with the P100 GPU for 60 epochs starting from the base YOLOv11s.pt

- I ran the training script, which also contained augmentation of about 10 degrees to reduce overfitting and obtained the best.pt weight .

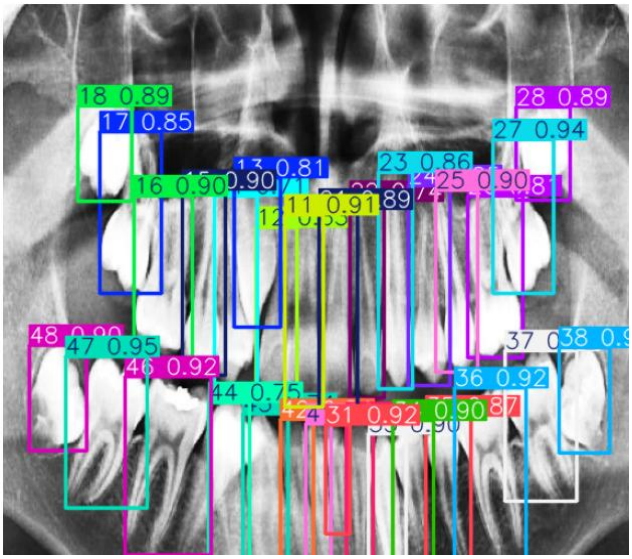
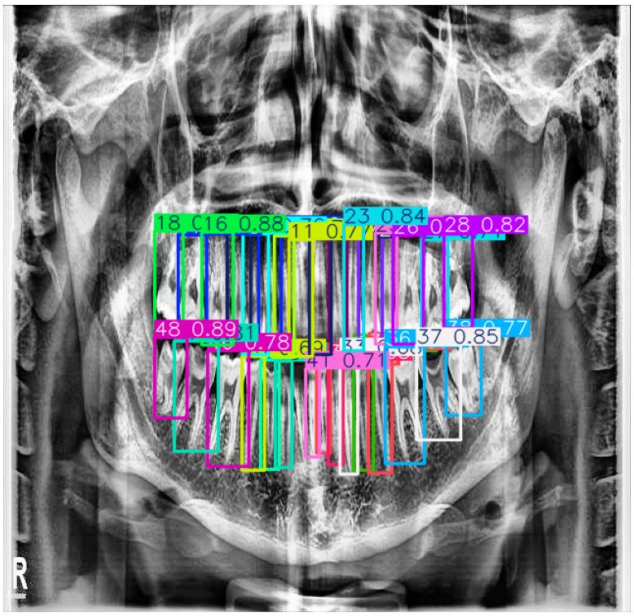
- Then I performed inference, during which I applied Non-Maximum Suppression (NMS) to filter out overlapping bounding boxes. This ensured the model retained only the single highest-confidence prediction per physical tooth, eliminating duplicate labels.

- For choosing a test image, I made sure to check scans with all 32 teeth to ensure that the model was properly working.

- This approach successfully yielded a 1-to-1 mapping during inference. The model correctly identified and numbered all 32 teeth in the test scans without duplicates, achieving mAP50 of about 88.9% despite the small dataset size.

Results

```
Validating /kaggle/working/runs/detect/train/weights/best.pt...
Ultralytics 8.4.23 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla P100-PCIE-16GB, 16269MiB)
YOLO11s summary (fused): 101 layers, 9,425,184 parameters, 0 gradients, 21.4 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 4/4 1.1s/it 4.5s1.3s
                   all        100       2697      0.841      0.871      0.889      0.599
                    13         88         88       0.86      0.907      0.876      0.559
                    23         86         86      0.799      0.895      0.886      0.588
                    33         96         96      0.833      0.854      0.832      0.533
                    43         93         93      0.814      0.849      0.835       0.56
                    21         87         87      0.892      0.857      0.937      0.623
                    41         93         93      0.768      0.712      0.825      0.444
                    31         92         92      0.867      0.709      0.846      0.471
                    11         88         88      0.851      0.909       0.92      0.615
                    16         84         84      0.848       0.93      0.948      0.696
                    26         76         76      0.893      0.882      0.922      0.608
                    36         75         75      0.903       0.92      0.928      0.747
                    46         74         74       0.83      0.932      0.898      0.714
                    14         80         80       0.76        0.9      0.891      0.519
                    34         89         89      0.846      0.863      0.886      0.598
                    44         89         89      0.855      0.862      0.874      0.602
                    24         80         80       0.79       0.75      0.839      0.488
                    22         86         86       0.92      0.798      0.885      0.576
                    32         92         92      0.801      0.837      0.849      0.531
                    42         93         93      0.804      0.817      0.813      0.453
                    12         86         86      0.885      0.884      0.877      0.542
                    17         85         85      0.822      0.953      0.941      0.675
                    27         90         90      0.856      0.867      0.897      0.596
                    37         82         82      0.854      0.927       0.92      0.713
                    47         81         81      0.852      0.926      0.909       0.75
                    15         85         85      0.873      0.892      0.897      0.565
                    25         83         83      0.884      0.828       0.87      0.563
                    35         82         82      0.827      0.874      0.903      0.639
                    45         84         84      0.783      0.948      0.912      0.646
                    18         70         70      0.805      0.929      0.911      0.576
                    28         77         77       0.89      0.837      0.934      0.592
                    38         77         77      0.837      0.909      0.886      0.665
                    48         74         74      0.803      0.919        0.9       0.73
```

Key Observations from the Output:

- **Complete Detection:** The YOLO model successfully located all individual teeth within the scan without missing any targets.
- **Perfect 1-to-1 Mapping:** The custom data.yaml translation layer successfully resolved the dataset's labeling error. The model correctly applied the spatial FDI notation (Quadrants 1 through 4) across the entire jawline.
- **No Duplication:** Thanks to the application of Non-Maximum Suppression (NMS), there are no overlapping bounding boxes or conflicting labels on a single tooth (e.g., no double predictions of tooth 17). The bounding boxes are tight and accurately distinct.